THE GYSIN MAP OF A MORPHISM OF SMOOTH SCHEMES OVER A FIELD IS MULTIPLICATIVE AND EXISTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FFB (sources/stacks-chow/pinned-original-source.tex line 14437).

Allowed standard prerequisites (editorial).

- Existence of the Gysin map for a local complete intersection morphism. Same chapter, Definition labelled definition-lci-gysin, cited at proof lines 14445-14448. Shows f^! exists via the immersion X -> X x_k Y.
- Identification of Chow classes of smooth schemes with caps of bivariant classes. Same chapter, Lemma labelled lemma-identify-chow-for-smooth, cited at proof lines 14456-14462 and 14470-14476. Writes alpha and beta as caps of bivariant classes against the fundamental class.
- The value of f^! on the fundamental class and its commutation with bivariant classes. Same chapter, Lemma labelled lemma-lci-gysin-easy, cited at proof lines 14454-14456, and Lemma labelled lemma-lci-gysin-commutes, cited at proof lines 14464-14468. Supplies f^![Y]_e = [X]_d and the commutation used throughout.
- Composition of Gysin maps and products. Same chapter, Lemma labelled lemma-lci-gysin-composition, cited at proof lines 14478-14479. Underlies the alternative proof through the product morphism.

Source reference. Original definition 0FF3: sources/stacks-chow/pinned-original-source.tex, source lines 13530–13530. Same-chapter pointer into the author environment definition at source line 13530; invoked at proof line 14446 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF7: sources/stacks-chow/pinned-original-source.tex, source lines 13643–13643. Same-chapter pointer into the author environment lemma at source line 13643; invoked at proof line 14451 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0FC2: sources/stacks-chow/pinned-original-source.tex, source lines 14403–14403. Same-chapter pointer into the author environment lemma at source line 14403; invoked at proof line 14454 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF6: sources/stacks-chow/pinned-original-source.tex, source lines 13613–13613. Same-chapter pointer into the author environment lemma at source line 13613; invoked at proof line 14455 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF5: sources/stacks-chow/pinned-original-source.tex, source lines 13575–13575. Same-chapter pointer into the author environment lemma at source line 13575; invoked at proof line 14478 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let k be a field. Let $f : X \to Y$ be a morphism of schemes smooth over k. Then the Gysin map exists for f and $f^!(\alpha \cdot \beta) = f^!\alpha \cdot f^!\beta$.*

Proof. Observe that $X \to X \times_k Y$ is an immersion of X into a scheme smooth over Y. Hence the Gysin map exists for f (Definition 0FF3). To prove the formula we may decompose X and Y into their connected components, hence we may assume X is smooth over k and equidimensional of dimension d and Y is smooth over k and equidimensional of dimension e. Observe that $f^![Y]_e = [X]_d$ (see for example Lemma 0FF7). Write $\alpha = c \cap [Y]_e$ and $\beta = c' \cap [Y]_e$ and hence $\alpha \cdot \beta = c \cap c' \cap [Y]_e$, see Lemma 0FC2. By Lemma 0FF6 we know that $f^!$ commutes with both c and c'. Hence

$$\begin{aligned} f^!(\alpha \cdot \beta) &= f^!(c \cap c' \cap [Y]_e) \\ &= c \cap c' \cap f^![Y]_e \\ &= c \cap c' \cap [X]_d \\ &= (c \cap [X]_d) \cdot (c' \cap [X]_d) \\ &= (c \cap f^![Y]_e) \cdot (c' \cap f^![Y]_e) \\ &= f^!(\alpha) \cdot f^!(\beta) \end{aligned}$$

as desired where we have used Lemma 0FC2 for X as well.

An alternative proof can be given by proving that $(f \times f)^!(\alpha \times \beta) = f^!\alpha \times f^!\beta$ and using Lemma 0FF5. □